

GROUND FLOOR PLAN

ADMINISTRATION

Ton = 2.6

Cfm=1285

No of Diffusers = 1

Diffuser Size = 600 x 600 mm

FLXØ550mm Type

OFFICE

Ton = 0.82

Cfm=404

No of Diffusers = 1

Diffuser Size = 300 x 300 mm

FLXØ250mm Type

DIRECTOR / CEO

Ton = 1.42

Cfm=694

No of Diffusers = 2 = $694/2 = 347$

Diffuser Size = 300 x 300 mm

FLXØ250mm Type

OFFICE ACCOUNT

Ton = 1.274

Cfm=623

No of Diffusers = 2 = $625/2 = 312$

Diffuser Size = 300 x 300 mm

FLXØ250mm Type

OFFICE 01

Ton = 0.3

Cfm=153

Ceiling Diffusers

Product: SAD Shape: Square Usage: Supply Unit System: Imperial

NC max: 35 CFM: 1285

Vn max: 500 fpm

Velocity: Air Terminal Velocity: 100 fpm

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Size		24x24	
An	4 ft2	Vn	321.25 fpm
Ak	1.4 ft2	Vk	887.375 fpm
Throw	28.7 ft	Pt	0.04 inchH ₂ O
		NC	32

I.S.V. = Insignificantly Small Value

Conditions:

- * Supply
- * Damper is fully open.
- * Noise Criteria values are based on (10 dB) room attenuation
- * For Rectangular Diffusers- throw values mentioned are for the longer side of the diffuser- for shorter sides throw values are 0.7-0.75 of the mentioned ones.

Throw Correction

3 way	+ 10 - 20 %
2 way	+ 20 - 30 %
1 way	+ 40 - 50 %

No of Diffusers = 1

Diffuser Size = 300 x 300 mm

FLX250mm Type

MEETING

Ton = 2.143

Cfm=1048

No of Diffusers = 3 = 1048/3 = 349

Diffuser Size = 300 x 300 mm

FLXØ250mm Type

RESTAURANT

Ton = 13.6

Cfm=6164

No of Diffusers = 4 = 6164/5 = 1233

Diffuser Size = 600 x 600 mm

FLXØ550mm Type

VIP LOUNGE

Ton = 6.6

Cfm=3356

No of Diffusers = 2 = 3356/2 = 1678

Diffuser Size = 600 x 600 mm

FLXØ550mm Type

VIP CHAMBER 01

Ton = 1.2

Cfm=602

No of Diffusers = 1

Diffuser Size = 450 x 450 mm

FLXØ400mm Type

VIP CHAMBER 02

Ton = 1.7

Ceiling Diffusers

Product

Shape

Usage

Unit System

SAD

Square

Supply

Imperial

NC max

45

CFM

1678

Vn max

500

fpm

Velocity

Air Terminal Velocity

100

fpm

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Size

24x24

An

4

R2

Vn

419.5

fpm

Ak

1.4

R2

Vk

1158.625

fpm

Throw

38.17

ft

Pt

0.068

inchW/G

NC

41

I.S.V. = Insignificantly Small Value

Conditions :

Throw Correction

* Supply

3 way + 10 - 20 %

* Damper is fully open.

2 way + 20 - 30 %

* Noise Criteria values are based on (10 dB) room attenuation

1 way + 40 - 50 %

* For Rectangular Diffusers- throw values mentioned are for the longer side of the diffuser- for shorter sides throw values are 0.7-0.75 of the mentioned ones.

Cfm=870

No of Diffusers = 1

Diffuser Size = 450 x 450 mm

FLXØ400mm Type

WC NORTH

EX-CFM = 50

No of Disc Valve = 3

Total Passage Area = 100 ft²

No of Disc Valve = 2

Each Disc Valve Cfm = 50

Total EX-CFM = 150+100 = 250

FLXØ150mm Type

WC SOUTH

EX-CFM = 50

No of Disc Valve = 3

Total Passage Area = 119 ft²

No of Disc Valve = 2

Each Disc Valve Cfm = 60

Total EX-CFM = 150+120 = 270

FLXØ150mm Type

MALE WC

EX-CFM = 50

No of Disc Valve = 1

Total Passage Area = 118 ft²

No of Disc Valve = 2

Each Disc Valve Cfm = 60

Total EX-CFM = 50+120 = 170

FLXØ150mm Type

FEMALE WC

EX-CFM = 50

No of Disc Valve = 5

Total Passage Area = 60 ft²

No of Disc Valve = 1

Each Disc Valve Cfm = 60

Total EX-CFM = 250+60 = 310

FLX150mm Type

GRAND TOTAL EX-CFM = 170+310 = 480

KITCHEN HOOD VENTILATION

Kitchen Area = 5.7m x 6.8m

Ex-Cfm = 1200 X 990/1000 = 1188

Ex Duct Size = 425W x 300H

Total Ex-Cfm = 2 x Ex-Cfm = 2 x 1188 = 2376

Main Duct Size = 550W x 400H

FRESH AIR = 80%

Fresh Air Cfm = 0.8 x 2376 = 1900

FA Duct Size = 450W x 400H

Grill Size = 2000L x 200H x 40T (8-inch Height)

FIRST FLOOR PLAN

MAIN HALL

Ton = 75.5

Cfm=31063

Zone 01 = Main Hall & Excluding Balcony

Zone 02 = Below Balcony Area & Projection Room(278Cfm)

No of Machines = 2 AHU (Each 32.08 Tr, it covers main hall)

= 2 FCU (FCU04A 5.7 Tr & FCU04B 5.8 Tr, it covers below Balcony area & Projection room)

Zone 01

No of Supply Diffuser & Grill = 28 = 30785/28 = 1100

Diffuser Size = 550 x 550mm

Linear Supply Grill Size = 1000L x 250H 50W mm

Linear Return Grill Size = 1500L x 250H 50W mm (18nos of Each Cfm = 1710)

Zone 02

FCU04A = 2200 Cfm

FCU04B = 2487 Cfm

No of Grill = 6 = 4400/6 = 1100

Diffuser Size = 550 x 550mm

Ceiling Diffusers

Product	Shape	Usage	Unit System
SAD	Square	Supply	Imperial

NC max: 35 CFM: 1100

Vn max: 500 fpm

Velocity: Air Terminal Velocity: 100 fpm

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Size		21x21	
An	3.063 ft2	Vn	359.677 fpm
Ak	1.065 ft2	Vk	1002.097 fpm
Throw	27.968 ft	Pt	0.049 inchWG
		NC	33

I.S.V. = Insignificantly Small Value

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Throw Correction

3 way + 10 - 20 %
2 way + 20 - 30 %
1 way + 40 - 50 %


Ceiling Diffusers

Product

SAD

Shape

Square

Usage

Supply

Unit System

Imperial

NC max

35

CFM

1098

Vn max

500

fpm

Velocity

Air Terminal Velocity

100

▼

fpm

Find Nearest Match >>

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Size	21x21			
An	3.063	ft2	Vn	359.032 fpm
Ak	1.065	ft2	Vk	1000.29 fpm
			Pt	0.049 inchWG
Throw	27.903	ft	NC	33

I.S.V. = Insignificantly Small Value

Conditions :

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- * Damper is fully open.
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Throw Correction

3 way + 10 - 20 %
2 way + 20 - 30 %
1 way + 40 - 50 %

FEMALE WC

EX-CFM = 50

No of Disc Valve = 6

Total Passage Area = 125 ft²

No of Disc Valve = 3

Each Disc Valve Cfm = 43

Total EX-CFM = 400+125 = 525

FLX150mm Type

GRAND TOTAL EX-CFM = 538+525 = 1063

MALE WC

EX-CFM = 50

No of Disc Valve = 2

Total Passage Area = 119 ft²

No of Disc Valve = 2

Each Disc Valve Cfm = 60

Total EX-CFM = 100+120= 220

FLXØ150mm Type

FEMALE WC

EX-CFM = 50

No of Disc Valve = 6

Total Passage Area = 174 ft²

No of Disc Valve = 2

Each Disc Valve Cfm = 87

Total EX-CFM = 400+174 = 474

FLX150mm Type

GRAND TOTAL EX-CFM = 219+474 = 693

SECOND FLOOR PLAN

BALCONY

Ton = 30.1

Cfm=10612

FLXØ550mm Type

MALE WC

EX-CFM = 50

No of Disc Valve = 2

Total Passage Area = 170 ft²

No of Disc Valve = 2

Each Disc Valve Cfm = 85

Total EX-CFM = 100+170= 270

FLXØ150mm Type

FEMALE WC

EX-CFM = 50

No of Disc Valve = 4

Total Passage Area = 125 ft²

No of Disc Valve = 2

Each Disc Valve Cfm = 62.5

Total EX-CFM = 200+125 = 325

FLX150mm Type

GRAND TOTAL EX-CFM = 270+325 = 595


Ceiling Diffusers

Product

SAD

Shape

Square

Usage

Supply

Unit System

Imperial

NC max

35

CFM

1327

Vn max

500

fpm

Velocity

Air Terminal Velocity

100

fpm

Find Nearest Match >>

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Exit

Size

24x24

An

4

ft2

Vn

331.75

fpm

Ak

1.4

ft2

Vk

915.725

fpm

Pt

0.043

inchWG

Throw

29.54

ft

NC

33

I.S.V. = Insignificantly Small Value

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2 way + 20 - 30 %

1 way + 40 - 50 %